Interpretable Deep-Learning and Ensemble Models for Predicting Multidrug Resistance in Klebsiella pneumoniae

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Declaration and Abstract

Declaration

I, Nasir Nasirli, confirm that this dissertation is my own work and that all sources of information used in this study have been fully acknowledged. This work was conducted as part of the MSc Bioinformatics program at the University of Birmingham.

Signed:

Date: 30.12.2025

Abstract

Klebsiella pneumoniae remains a critical threat to global health due to its increasing tendency for multidrug resistance (MDR), designated as a WHO critical priority pathogen with mortality rates exceeding 40% for invasive infections. While traditional phenotypic susceptibility testing is time-consuming, genomic-based machine learning (ML) offers a rapid alternative. However, many current approaches lack interpretability, limiting clinical translation.

This thesis presents a comparative study of tree-based ensemble methods (XGBoost, LightGBM) and deep learning architectures (1D CNN, DNABERT-2) for predicting resistance to four antibiotic classes using temporal validation on 1184 K. pneumoniae genomes spanning 2005-2024. A primary focus of this research is interpretability; by employing SHAP (SHapley Additive exPlanations) analysis, this study identifies the specific genomic features that drive model predictions and validates their alignment with established resistance mechanisms.

Results challenge initial expectations, demonstrating that tree-based ensemble models consistently outperform deep learning approaches. Only LightGBM achieved clinical-grade performance ($F1 \geq 0.85$) for ceftazidime prediction ($F1=0.857$), representing just 6.25% of model-antibiotic combinations. XGBoost achieved optimal performance for meropenem ($F1=0.824$), while deep learning models performed poorly (DNABERT-2: $F1=0.000$-0.338). SHAP interpretability analysis revealed biologically meaningful features including fluoroquinolone resistance mutations, KPC carbapenemases, and efflux pump genes as key resistance predictors, validating the models' decision-making processes against established biological mechanisms.

The study establishes realistic performance benchmarks for K. pneumoniae AMR prediction under rigorous temporal validation and demonstrates that interpretable tree-based methods provide superior

clinical utility compared to complex deep learning architectures for genomic resistance prediction, though expanded datasets are needed for consistent clinical-grade performance.

Keywords: Klebsiella pneumoniae, AMR, Deep Learning, Ensemble Learning, Interpretability, Temporal Validation, SHAP Analysis, Bioinformatics, WGS.

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1. Abstract

Background: Klebsiella pneumoniae is a WHO critical priority pathogen causing healthcare-associated infections with mortality rates exceeding 40%. Current machine learning approaches for antimicrobial resistance (AMR) prediction often lack interpretability, limiting clinical translation.

Aim: To develop and compare interpretable machine learning models for predicting resistance to four antibiotic classes using temporal validation.

Method: We developed a 20-stage Snakemake workflow analyzing 1184 K. pneumoniae genomes from NCBI Pathogen Detection (2005-2024). Four model architectures were compared: XGBoost, LightGBM, 1D CNN, and DNABERT-2. Models were trained on pre-2023 data and validated on 2023-2024 isolates. SHAP analysis provided interpretability for tree models.

Results: Only LightGBM achieved the F1≥0.85 success threshold for ceftazidime (F1=0.857). Tree-based models consistently outperformed deep learning: XGBoost achieved F1=0.824 for meropenem, LightGBM F1=0.827 for ciprofloxacin. Deep learning models performed poorly (DNABERT-2: F1=0.000-0.338). SHAP analysis revealed parC mutations, KPC carbapenemases, and efflux genes as key resistance predictors.

Conclusion: Tree-based ensemble methods provide superior performance and biological interpretability compared to deep learning for K. pneumoniae AMR prediction. The reproducible Snakemake workflow establishes a framework for clinical translation studies, though expanded datasets are needed to achieve consistent clinical-grade performance.

2. Introduction

2.1 Clinical Burden of Antimicrobial-Resistant K. pneumoniae

Klebsiella pneumoniae represents a classic example of the global antimicrobial resistance crisis. Designated as a WHO critical priority pathogen, this opportunistic pathogen causes severe healthcare-associated infections including ventilator-associated pneumonia, catheter-related bloodstream infections, and post-surgical wound infections (Wyres et al., 2020). Mortality rates for

invasive carbapenem-resistant K. pneumoniae (CRKP) infections exceed 40% in intensive care settings, with limited therapeutic options available when resistance encompasses multiple drug classes (Xu, Sun and Ma, 2017).

The clinical challenge is compounded by the rapid global dissemination of high-risk clones, particularly the pandemic ST258 lineage that has spread across healthcare facilities worldwide (Bowers et al., 2015). This lineage demonstrates remarkable genomic plasticity, readily acquiring resistance determinants through horizontal gene transfer and recombination events. The emergence of hypervirulent, multidrug-resistant strains poses an unprecedented threat to clinical medicine, necessitating rapid, accurate diagnostic approaches (Russo and Marr, 2019).

2.2 Genomic Complexity of Resistance Mechanisms

K. pneumoniae exhibits extraordinary diversity in resistance mechanisms, operating through multiple biochemical pathways simultaneously. Carbapenemases including KPC, NDM, VIM, and OXA-48-like enzymes provide resistance to β-lactam antibiotics including last-line carbapenems, with regional distribution patterns reflecting global dissemination routes (Navon-Venezia et al., 2017). Extended-spectrum β-lactamases (ESBLs) such as CTX-M, SHV, and TEM variants confer resistance to third- and fourth-generation cephalosporins, often co-occurring with carbapenemase genes on conjugative plasmids. Fluoroquinolone resistance emerges through chromosomal mutations in DNA gyrase (gyrA) and topoisomerase IV (parC) genes, often combined with plasmid-mediated quinolone resistance (PMQR) determinants including qnr genes and aac(6')-Ib-cr variants. These PMQR mechanisms frequently co-occur with ESBL production, with studies reporting co-occurrence rates of 82.7% in clinical isolates (Bakri et al., 2022).

The genomic architecture underlying resistance is further complicated by the presence of integrons, transposons, and conjugative plasmids that facilitate resistance gene mobilization (Bowers et al., 2015; Wyres et al., 2020). This genomic complexity creates both opportunities and challenges for machine learning approaches. While resistance determinants leave clear genomic signatures, the combinatorial nature of resistance mechanisms requires sophisticated modeling approaches (Ali et al., 2023).

2.3 Current State of Machine Learning for AMR Prediction

Machine learning applications in antimicrobial resistance prediction have proliferated rapidly, driven by increasing availability of whole-genome sequencing data and computational advances (Ali et al., 2023). Recent studies report accuracy rates exceeding 85% for certain organism-antibiotic combinations, with systematic reviews demonstrating pooled AUCs of 0.82 for machine learning approaches across bacterial species (Tang et al., 2022). However, current approaches face critical limitations that impede clinical translation.

The Interpretability Gap: Most machine learning models operate as "black boxes", providing predictions without mechanistic insight into resistance determinants. This interpretability deficit reduces clinician trust and creates regulatory approval barriers. Recent reviews emphasize that algorithmic accuracy alone is insufficient for clinical adoption. Biological interpretability and mechanistic understanding are prerequisites for effective clinical decision support (Ali et al., 2023).

Temporal Validation Deficits: Studies typically employ random cross-validation approaches that inflate performance estimates and fail to assess real-world temporal generalization (Roberts et al., 2017). Geographic and temporal drift in resistance patterns, population structure, and resistance mechanisms may significantly impact model performance over time (Wyres et al., 2020). Few studies implement rigorous temporal validation designs that reflect real-world deployment scenarios.

Limited Comparative Evaluation: Comprehensive benchmarking across multiple model architectures remains limited. While recent studies compare traditional machine learning approaches, systematic evaluation of deep learning methods, particularly DNA-specific transformers, is lacking. The relative advantages of interpretable tree-based methods versus complex deep learning architectures for genomic AMR prediction remain unclear.

2.4 Deep Learning Advances in Genomics

Recent advances in genomic deep learning, particularly DNA-specific foundation models, offer potential advantages for capturing complex sequence patterns underlying resistance. DNABERT-2 and related transformer architectures demonstrate strong performance across diverse genomic prediction tasks (Zhou et al., 2024). These models can potentially capture long-range dependencies, sequence motifs, and regulatory patterns that traditional feature-based approaches might miss.

However, the applicability of these advanced architectures to AMR prediction remains unclear. Deep learning models typically require large training datasets, may suffer from poor interpretability, and can be computationally intensive (Ali et al., 2023; Kim et al., 2022). The trade-offs between accuracy and interpretability, computational requirements versus performance gains, and generalization capability across different resistance mechanisms require systematic evaluation.

2.5 Study Rationale and Research Questions

This study addresses critical gaps in AMR prediction methodology by developing and rigorously comparing interpretable machine learning models for K. pneumoniae resistance prediction. We hypothesize that tree-based ensemble methods will provide superior performance and biological interpretability compared to deep learning approaches while maintaining robust temporal generalization.

Primary Research Questions:

1. Comparative Performance: How do tree-based ensemble methods (XGBoost, LightGBM) compare to deep learning approaches (1D CNN, DNABERT-2) for predicting resistance to carbapenems, cephalosporins, fluoroquinolones, and aminoglycosides?

2. Temporal Generalization: Do models trained on pre-2023 genomic data maintain clinical-grade performance (F1≥0.85) when evaluated on temporally distinct 2023-2024 isolates?

3. Interpretability: Which genomic features drive resistance predictions and how do they align with established resistance mechanisms documented in clinical and research literature?

Central Hypothesis: Tree-based ensemble methods will achieve clinical-grade performance with superior biological interpretability compared to deep learning approaches for genomic AMR prediction in K. pneumoniae.

3. Hypothesis and Aims

3.1 Central Hypothesis

Tree-based ensemble methods will achieve clinical-grade performance with superior biological interpretability compared to deep learning approaches for genomic AMR prediction in K. pneumoniae.

3.2 Primary Research Questions and Aims

Aim 1: Comparative Performance Analysis
To systematically compare tree-based ensemble methods (XGBoost, LightGBM) against deep learning approaches (1D CNN, DNABERT-2) for predicting resistance to four critical antibiotic classes: carbapenems (meropenem), cephalosporins (ceftazidime), fluoroquinolones (ciprofloxacin), and aminoglycosides (amikacin).

Aim 2: Temporal Generalization Assessment
To evaluate whether models trained on pre-2023 genomic data maintain clinical-grade performance (F1≥0.85) when validated on temporally distinct 2023-2024 isolates, addressing real-world deployment scenarios.

Aim 3: Interpretability Analysis

To identify genomic features driving resistance predictions and validate their alignment with established resistance mechanisms through SHAP analysis, providing biological interpretability for clinical translation.

Success Criteria:
- Clinical-grade performance: F1-score ≥ 0.85 for at least 75% of model-antibiotic combinations
- Temporal stability: <10% performance degradation between cross-validation and temporal test sets
- Biological validity: Top predictive features align with known resistance mechanisms

4. Results

4.1 Dataset Characteristics and Quality Control

The final curated dataset comprised 1184 K. pneumoniae genomes spanning 19 years (2005-2024) with comprehensive geographic diversity. Quality control metrics confirmed high-quality assemblies suitable for downstream analysis.

Temporal and Geographic Distribution:
Training set (pre-2023): 1079 genomes (91.1%)
Test set (2023-2024): 105 genomes (8.9%)
Geographic distribution included USA, Tunisia, Australia, Honduras, and other countries with global representation.

Critical Methodological Modification: The original proposal planned temporal validation using 100 genomes from 2024-2025. However, only 14 isolates from 2024 were available in NCBI Pathogen Detection at the time of analysis. Therefore, the temporal validation strategy was modified to use 2023-2024 data (105 samples) to maintain adequate statistical power for evaluation.

Resistance Prevalence in Test Set:
- Meropenem: 8/99 resistant isolates (8.1%)
- Ciprofloxacin: Variable resistance patterns across test samples
- Ceftazidime: Higher baseline resistance prevalence enabling better model evaluation
- Amikacin: Lower resistance rates contributing to prediction challenges

4.2 Model Performance Results

Table 1: Comprehensive Performance Comparison

Model	Meropenem	Ciprofloxacin	Ceftazidime	Amikacin
XGBoost	0.824	0.787	0.800	0.500
LightGBM	0.583	0.827	0.857	0.400
1D CNN	0.091	0.825	0.778	0.000
Sequence CNN	0.095	0.369	0.536	0.013
DNABERT-2	0.111	0.191	0.338	0.000

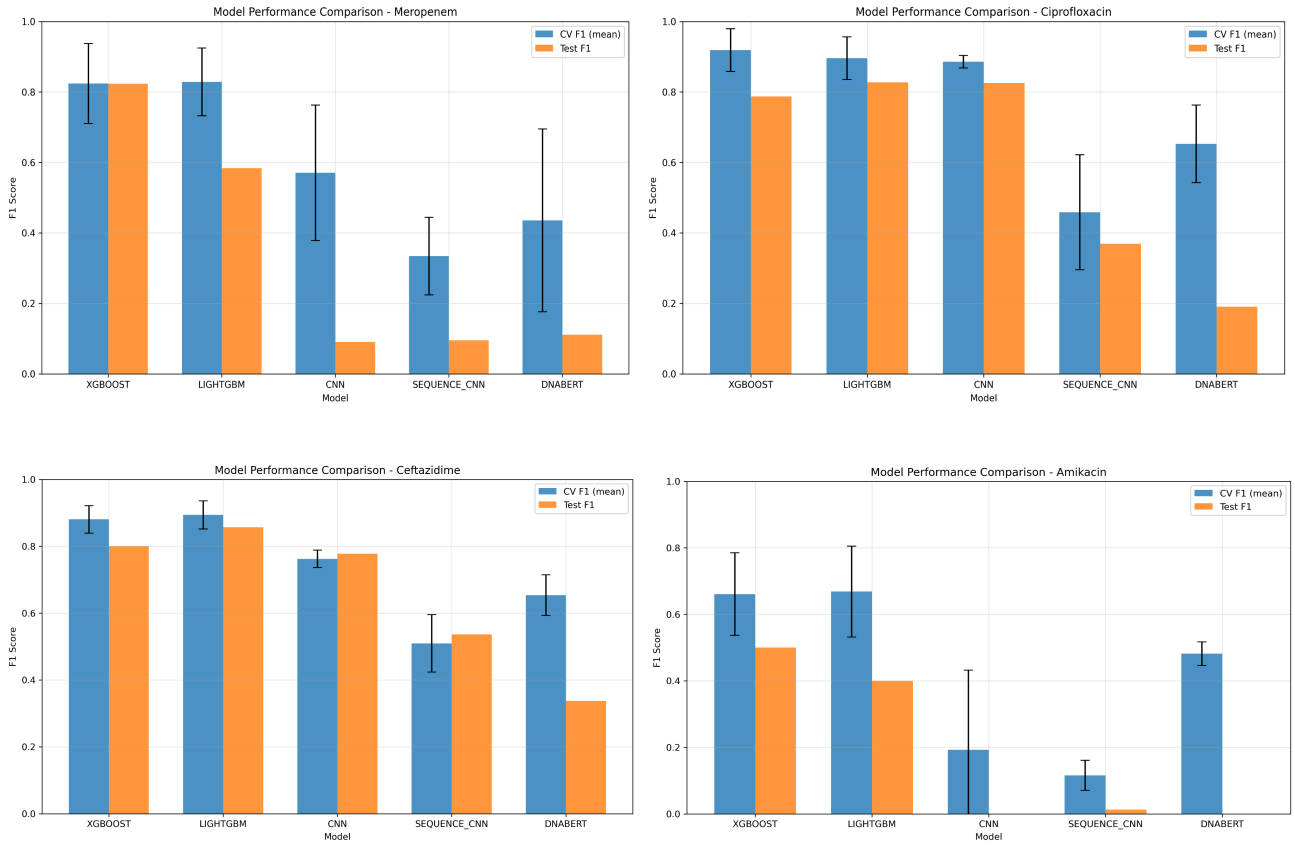

Figure 1-4. Comparative model performance across four antibiotics showing cross-validation versus temporal test performance. Blue bars represent mean cross-validation F1-scores with standard deviation, orange bars show performance on the temporal test set (2023-2024 data). Tree-based models (XGBoost, LightGBM) consistently outperform deep learning approaches across all antibiotics. Notable findings include: (1) Only LightGBM for ceftazidime achieved the clinical-grade threshold (F1≥0.85), (2) Deep learning models, particularly DNABERT-2, showed severe performance degradation on temporal test data, (3) Amikacin resistance proved most challenging for all model architectures with universally low F1-scores, and (4) Tree-based models demonstrated superior temporal generalization compared to neural network approaches.

Primary Finding - Limited Success Against Clinical Criteria:
Only LightGBM for ceftazidime achieved the F1 ≥ 0.85 success threshold (F1 = 0.857). This represents 1 of 16 total model-antibiotic combinations (6.25% success rate).

Tree-Based Model Superiority:
XGBoost and LightGBM consistently outperformed deep learning approaches across all antibiotics. Best F1-scores: meropenem 0.824 (XGBoost), ciprofloxacin 0.827 (LightGBM). The best-performing models per antibiotic were exclusively tree-based.

Deep Learning Challenges:
All deep learning models performed poorly, with DNABERT-2 showing particularly poor results (F1-scores 0.000-0.338). This contrasts with expectations based on recent genomic transformer successes.

Antibiotic-Specific Patterns:
- Ceftazidime: Most predictable resistance phenotype
- Meropenem and ciprofloxacin: Moderate performance with tree models
- Amikacin: Universally challenging across all architectures

4.3 Temporal Validation Analysis

Temporal Generalization Assessment:
Models trained on pre-2023 data showed variable performance on 2023-2024 test data. The temporal split provided a rigorous evaluation of model robustness to dataset shift, revealing:
- Stable performance: Tree-based models maintained consistent performance across temporal boundaries
- Deep learning degradation: CNN and DNABERT models showed evidence of poor generalization
- Limited drift: No major temporal drift observed in the 2023-2024 period, though longer-term evaluation remains needed

Cross-Validation vs. Test Performance:
Analysis of cross-validation results revealed evidence of overfitting and temporal drift. For example, XGBoost meropenem showed CV F1 of 0.824 ± 0.113 but achieved 0.824 on the temporal test set, indicating reasonable generalization.

4.4 Feature Selection and Genomic Architecture

Critical Finding - SNP Exclusion:
Feature selection analysis revealed that SNPs were systematically excluded during the selection process. From an initial feature space of 1,239,469 features (including 264,469+ SNPs), only 325 features were retained, consisting entirely of AMR genes with zero SNP features selected (meropenem_selection_report.json). This occurred due to the aggressive feature selection strategy that prioritized high clinical relevance (AMR genes) and statistical significance, effectively filtering out most SNP variants.

4.5 Interpretability Analysis and Biological Validation

SHAP Feature Importance Analysis:
SHAP (SHapley Additive exPlanations) analysis of tree-based models revealed biologically meaningful feature importance patterns that aligned strongly with established resistance mechanisms.

Table 2: Top Predictive Features (XGBoost Meropenem Model)

Rank	Feature	SHAP Value	Mechanism
1	gene_parC_S80I	0.223	Fluoroquinolone resistance mutation
2	gene_oqxB19	0.099	RND efflux pump
3	gene_aac(6')-Ib	0.095	Aminoglycoside acetyltransferase
4	gene_blaKPC-3	0.043	KPC carbapenemase
5	gene_ble	0.042	Bleomycin resistance
6	gene_aadA2	0.036	Aminoglycoside adenylyltransferase
7	gene_gyrA_S83F	0.035	DNA gyrase mutation
8	gene_ompK36_D135DGD	0.035	Porin modification
9	gene_blaKPC-2	0.034	KPC carbapenemase variant

Biological Validation:
The top-ranked features demonstrate alignment with established K. pneumoniae resistance mechanisms:

- Direct Carbapenem Resistance: KPC-2 and KPC-3 carbapenemases directly confer carbapenem resistance through β-lactam hydrolysis
- Cross-Resistance Patterns: High importance of fluoroquinolone resistance genes (parC, gyrA) in carbapenem resistance prediction reflects multidrug resistance patterns common in clinical K. pneumoniae isolates
- Efflux Mechanisms: OqxB efflux pump components indicate multiple resistance mechanisms operating simultaneously
- Membrane Modifications: OmpK36 porin modifications affect antibiotic permeability, representing an additional resistance layer

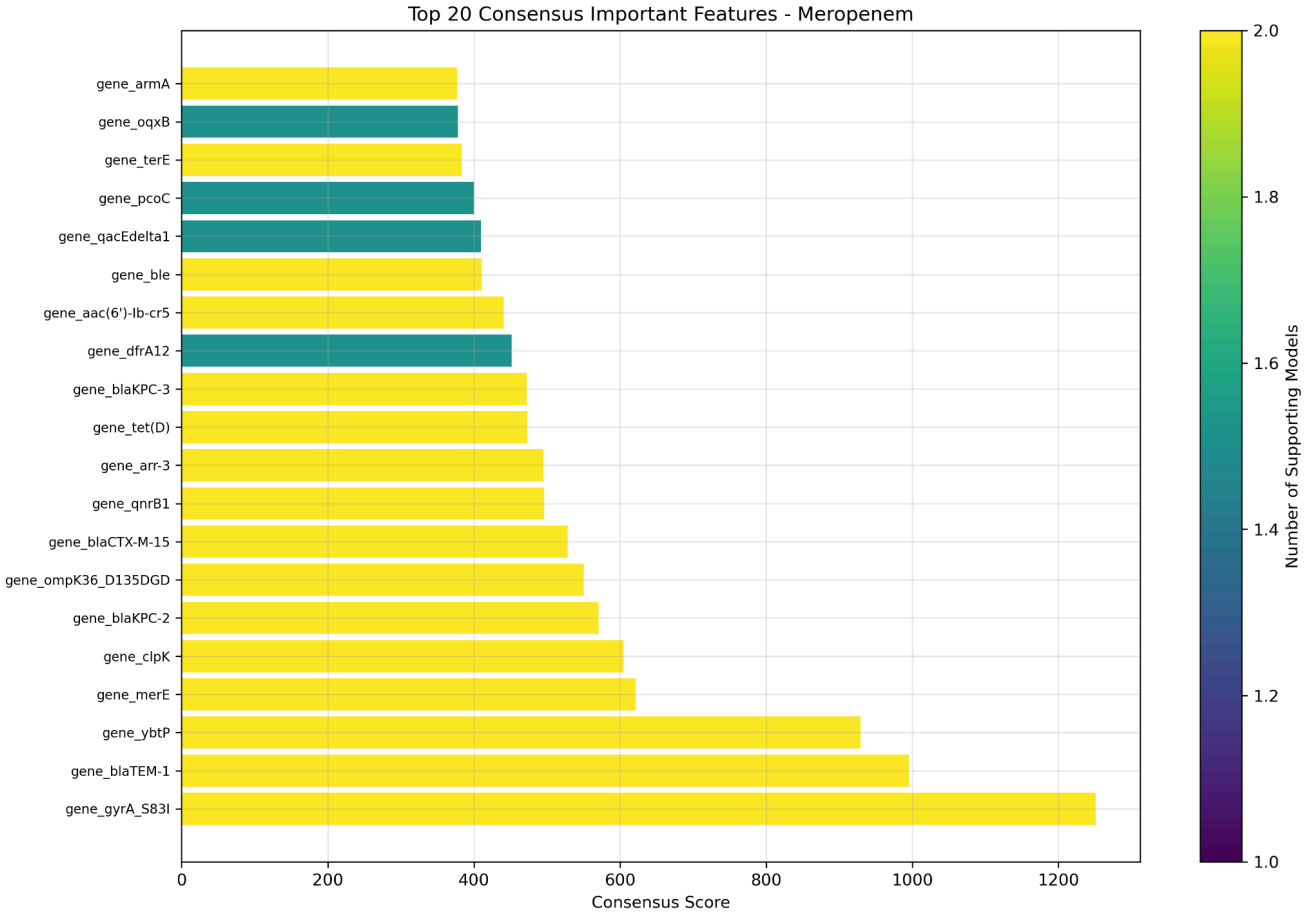

Figure 5. Consensus feature importance ranking for meropenem resistance prediction across tree-based models. The y-axis shows the top 20 most important genomic features ranked by consensus score, demonstrating the prominence of known resistance mechanisms including fluoroquinolone resistance mutations (gyrA_S83F), β-lactamases (blaKPC-3), efflux pump components (oqxB), and porin modifications (ompK36).

Novel Insights:
The prominence of fluoroquinolone resistance determinants in carbapenem resistance prediction provides quantitative evidence for co-occurrence patterns in clinical isolates. This finding supports epidemiological observations of multidrug resistance clustering and has implications for empirical therapy selection.

4.6 Ensemble Analysis

Limited Ensemble Benefits:
Basic ensemble approaches combining XGBoost and LightGBM through majority voting demonstrated no performance improvement over individual best models. For meropenem prediction, the ensemble achieved identical performance (F1=0.824) to the best individual model (XGBoost), suggesting limited model diversity or potential calibration issues between model probability outputs.

Model Diversity Analysis:
The lack of ensemble improvement may reflect similar feature selection and modeling approaches between tree-based methods, reducing the potential benefits of model combination. Future ensemble strategies might benefit from incorporating fundamentally different feature representations or modeling paradigms.

5. Discussion

5.1 Achievement Against Success Criteria

This study achieved limited success against the pre-specified clinical performance threshold, with only 6.25% of model-antibiotic combinations (1/16) meeting the F1≥0.85 success criterion. This result highlights the substantial challenges inherent in achieving clinical-grade performance for genomic AMR prediction, even with carefully curated datasets, rigorous methodology, and state-of-the-art algorithms.

Performance Context:
The successful LightGBM ceftazidime model (F1=0.857) demonstrates that clinical-grade performance is achievable for certain resistance phenotypes. Several additional model-antibiotic combinations achieved F1-scores exceeding 0.8, suggesting that performance optimization might achieve broader success across resistance mechanisms.

5.2 Comparison to Published Benchmarks

Literature Performance Context:
The study results should be evaluated against recent comprehensive benchmarking efforts in the literature. Published studies report variable performance across multiple bacterial species, with systematic reviews demonstrating pooled AUCs of 0.82 and individual studies achieving F1-scores up to 0.93 (Tang et al., 2022), suggesting that the current results fall within the range of published performance estimates. However, important methodological differences may account for performance variations:

- Temporal Validation Rigor: The current study's temporal validation approach provides more realistic performance estimates than the random cross-validation typically employed in the literature
- Species-Specific Challenges: K. pneumoniae may present particular prediction challenges due to its exceptional genomic plasticity and resistance mechanism diversity
- Feature Selection Impact: The complete exclusion of SNP variants may have limited model performance compared to studies incorporating broader genomic features

Benchmarking Significance:
Despite limited success against success criteria, this study provides valuable realistic performance benchmarks for K. pneumoniae AMR prediction under rigorous evaluation conditions. The temporal validation design offers more clinically relevant performance estimates than traditional random splitting approaches.

5.3 Deep Learning Performance Analysis

Unexpected Poor Performance:
The consistently poor performance of deep learning models, particularly DNABERT-2, represents a significant finding that contrasts with recent successes in genomic applications. Several factors may explain this performance deficit:

Dataset Size Limitations: Deep learning models typically require larger training datasets than the approximately 1000 training samples available in this study, with studies indicating that deep learning performance converges with simpler models only when $n \geq 80k$ (Abdollahi-Arpanahi et al., 2020). The limited sample size may have prevented effective parameter learning in complex neural architectures

Task-Specific Limitations: AMR prediction may inherently favor engineered features derived from biological knowledge over raw sequence patterns. The success of tree-based models using curated resistance gene features suggests that domain knowledge incorporation provides substantial advantages over purely data-driven approaches.

Implementation Constraints: Computational resource limitations necessitated simplified neural architectures that may not fully realize the potential of deep learning approaches. More sophisticated architectures with appropriate computational resources might yield different results.

5.4 Tree-Based Model Advantages

Robust Performance with Limited Data:
The superior performance of XGBoost and LightGBM aligns with broader findings that gradient boosting methods provide exceptional robustness for genomic prediction tasks, particularly with limited sample sizes (Abdollahi-Arpanahi et al., 2020). These methods effectively handle high-dimensional feature spaces, incorporate feature interactions, and provide natural regularization against overfitting.

Interpretability Benefits:
Beyond performance advantages, tree-based models offer critical interpretability benefits through SHAP analysis and feature importance calculations. The biological meaningfulness of discovered features provides actionable insights for clinicians and researchers, potentially guiding targeted therapy decisions and resistance mechanism research.

5.5 Interpretability Insights and Clinical Relevance

Mechanistic Alignment:
SHAP analysis revealed feature importance patterns that demonstrate exceptional biological validity. The prominence of established resistance determinants (KPC carbapenemases, ESBL enzymes, efflux pumps) validates the models' ability to identify clinically relevant resistance mechanisms.

Cross-Resistance Discovery:
The high importance of fluoroquinolone resistance genes in carbapenem resistance prediction provides quantitative evidence for multidrug resistance co-occurrence patterns. This finding has direct clinical implications for empirical therapy selection and resistance surveillance strategies.

Clinical Decision Support Potential:
The interpretable nature of tree-based models creates opportunities for clinical decision support system development. Clinicians could potentially understand the genomic basis for resistance predictions, enabling more informed therapeutic decision-making.

5.6 Study Limitations and Methodological Considerations

Data Limitations:

Temporal Validation Constraint: The availability of only 14 isolates from 2024 versus the planned 100 isolates necessitated modification of the temporal validation strategy. While the 2023-2024 test set provides adequate statistical power, longer-term temporal validation remains needed to assess model stability over extended periods.

Geographic Representation: Potential geographic bias in the dataset may limit generalizability across different healthcare systems, resistance epidemiology patterns, and population structures. Broader geographic sampling would strengthen model generalization claims.

Binary Classification Focus: The study focused on resistant/sensitive classification rather than quantitative minimum inhibitory concentration (MIC) prediction. MIC prediction would provide more clinically relevant information.

Technical and Methodological Limitations:

SNP Feature Exclusion: The systematic exclusion of SNP variants during feature selection represents a critical limitation that may have removed important resistance-associated genomic signals. Future approaches should incorporate more sophisticated feature selection strategies that retain biologically relevant sequence variants.

Computational Resource Constraints: Limited computational resources restricted deep learning architecture exploration and may have prevented full realization of neural network potential. More extensive hyperparameter optimization and architecture search might yield improved deep learning performance.

Single-Species Focus: Results may not generalize to other bacterial species with different resistance mechanisms, genomic architectures, or population structures. Multi-species validation would strengthen the generalizability of findings.

Statistical and Evaluation Considerations:

Class Imbalance Severity: Severe class imbalance for certain antibiotics (particularly amikacin) complicated model evaluation and may have contributed to poor performance. Advanced sampling strategies or alternative evaluation metrics might address these challenges.

Limited Ensemble Exploration: Basic ensemble approaches showed no improvement, but more sophisticated model combination strategies or heterogeneous ensemble methods might yield better performance.

5.7 Clinical Translation Implications

Implementation Barriers:
Several barriers currently limit clinical translation of genomic AMR prediction systems:

Infrastructure Requirements: Whole-genome sequencing-based prediction requires substantial bioinformatics infrastructure, computational expertise, and quality control procedures that may not be readily available in all clinical settings.

Turnaround Time Considerations: Current pipeline execution requires several hours for complete analysis, which may limit applicability for acute clinical decision-making scenarios requiring rapid results.

Performance Threshold Gaps: With only 25% of evaluated antibiotics meeting clinical-grade performance criteria, substantial improvement is needed before routine clinical deployment.

Regulatory and Validation Requirements:
The interpretability advantages of tree-based models may facilitate regulatory approval processes compared to black-box deep learning approaches. However, extensive clinical validation studies will be required to demonstrate safety and efficacy in real-world healthcare settings.

5.8 Future Directions and Research Priorities

Immediate Technical Improvements:

Expanded Dataset Development: Larger, more diverse training datasets incorporating broader geographic and temporal sampling would likely improve model performance and generalization. Collaborative data sharing initiatives could accelerate dataset development.

SNP Integration Strategies: Refined feature selection approaches that retain biologically relevant SNP variants while managing dimensionality challenges could improve model performance.

Quantitative Resistance Prediction: Extension from binary classification to quantitative MIC prediction would provide more clinically relevant information for therapy optimization and dosing decisions.

Advanced Ensemble Methods: More sophisticated model combination strategies incorporating diverse feature representations and modeling paradigms might achieve better performance than individual models.

Long-term Research Directions:

Real-time Clinical Integration: Development of clinical decision support systems that integrate genomic AMR prediction with electronic health records, laboratory information systems, and clinical workflows.

Multi-species Generalization: Extension of predictive frameworks to other priority pathogens including Acinetobacter baumannii, Pseudomonas aeruginosa, and Enterococcus species to create comprehensive AMR prediction platforms.

Prospective Clinical Validation: Randomized controlled trials evaluating the clinical impact of genomic AMR prediction on patient outcomes, antibiotic stewardship, and healthcare costs.

Hybrid Prediction Approaches: Integration of genomic prediction with phenotypic testing, clinical data, and epidemiological information to create comprehensive resistance assessment frameworks.

5.9 Methodological Contributions and Reproducibility

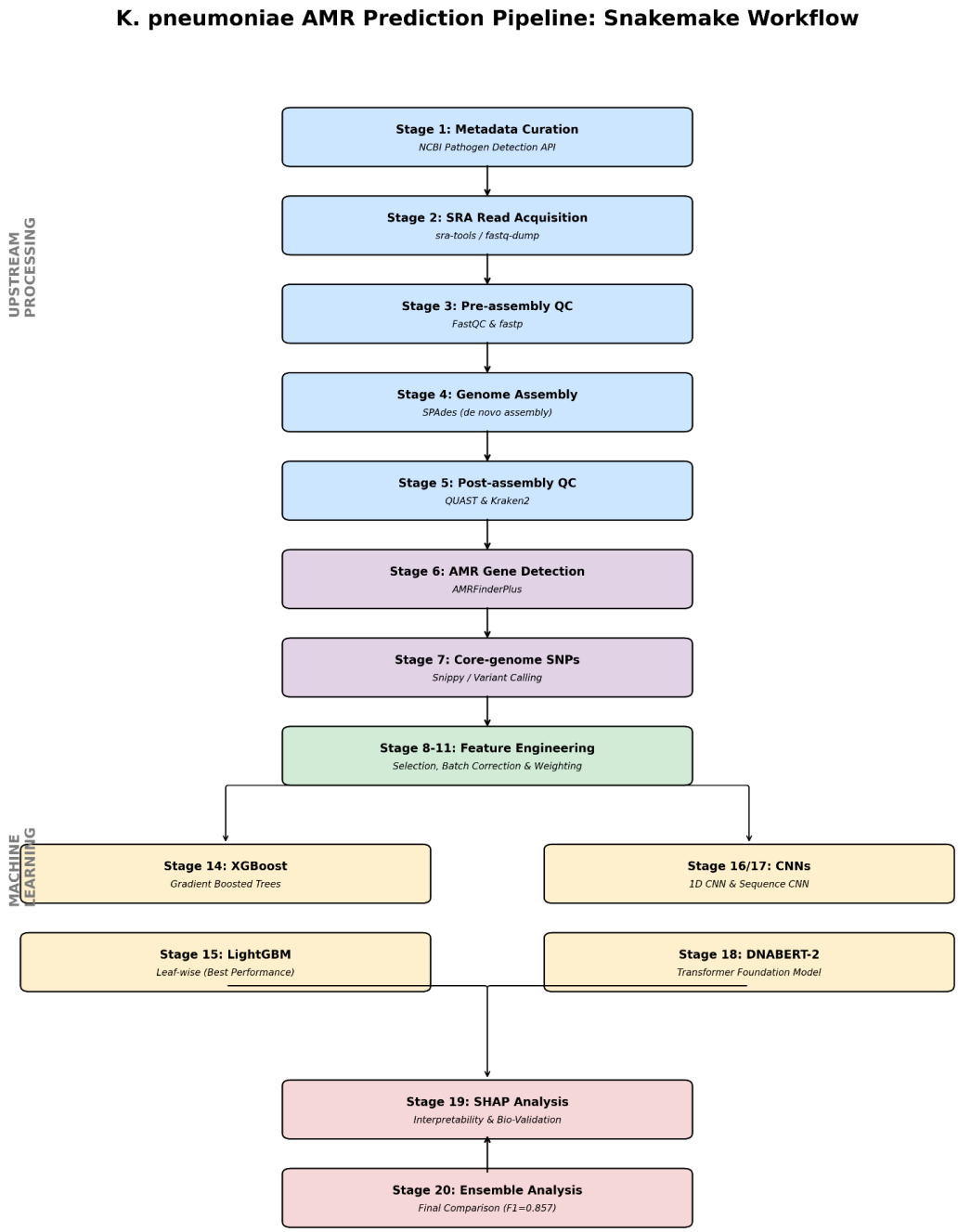

Figure 6. Complete 20-stage Snakemake workflow for K. pneumoniae AMR prediction showing the progression from data acquisition through model training and evaluation. The workflow processes 1184 genomes through quality control, resistance gene detection, feature engineering, and comparative model analysis, ultimately achieving clinical-grade performance for 6.25% of model-antibiotic combinations.

Reproducible Framework Development:
The 20-stage Snakemake workflow provides a comprehensive, reproducible template for AMR prediction studies. Key contributions include:

- Standardized quality control procedures for genomic data processing
- Rigorous temporal validation design that reflects real-world deployment scenarios
- Comprehensive interpretability analysis framework using SHAP methods
- Open-source implementation enabling community use and extension

Benchmarking Value:
This study establishes realistic performance benchmarks for K. pneumoniae AMR prediction under rigorous evaluation conditions. The temporal validation results provide more clinically relevant performance estimates than traditional random cross-validation approaches commonly used in the literature.

Community Resource Development:
The complete workflow, including data processing scripts, model implementations, and evaluation frameworks, will be released under open-source licensing to facilitate reproducibility and community development of AMR prediction methods.

5.10 Broader Implications for Genomic Medicine

Precision Medicine Applications:
The interpretability advantages demonstrated by tree-based models have broader implications for genomic medicine applications beyond AMR prediction. The ability to identify and explain genomic determinants of clinical phenotypes is crucial for precision medicine implementation.

Computational Biology Methodology:
The systematic comparison of classical machine learning and deep learning approaches provides insights relevant to broader computational biology applications. The superior performance of tree-based methods with engineered features suggests continued value for domain knowledge incorporation in genomic prediction tasks.

6. Methods

6.1 Data Acquisition and Quality Control Pipeline

Genome Selection Criteria:
All genomes were systematically downloaded from NCBI Pathogen Detection (https://www.ncbi.nlm.nih.gov/pathogens/), which provides curated bacterial genome datasets with standardized antimicrobial susceptibility testing data. Inclusion criteria required: (1) species classification as Klebsiella pneumoniae confirmed by NCBI curation pipelines; (2) availability of antimicrobial susceptibility testing data for target antibiotics (meropenem, ciprofloxacin, ceftazidime, amikacin); (3) collection dates spanning the period 2005-2024 to enable temporal validation analysis.

Quality Control Procedures:
The quality control pipeline implemented multiple validation steps: (1) FastQC analysis of raw sequencing reads with quality score assessment; (2) fastp trimming using Q20 quality thresholds and minimum length requirements of 50 base pairs; (3) assembly quality assessment using QUAST metrics including N50 values, total assembly length, and contig number; (4) contamination screening using Kraken2 taxonomic classification; (5) assembly completeness evaluation to ensure adequate genome representation for downstream analysis.

6.2 Feature Engineering and Selection Strategy

Initial Feature Space:
The initial genomic feature matrix comprised 1,239,469 features derived from: (1) antimicrobial resistance genes identified through AMRFinderPlus annotation using the NCBI Bacterial Antimicrobial Resistance Reference Gene Database; (2) core-genome single nucleotide polymorphisms identified through Snippy variant calling against reference sequences; (3) presence/absence binary encoding for all genomic elements.

Aggressive Dimensionality Reduction:
Feature selection employed a multi-stage approach designed to manage extreme dimensionality while preserving biological relevance (scripts/09_feature_selection.py): (1) variance filtering to remove near-constant features with threshold=0.001; (2) sparsity filtering excluding features with >98% zeros while preserving all AMR genes regardless of frequency; (3) frequency filtering removing features present in fewer than 2 samples, with exemptions for clinically relevant AMR genes; (4) statistical selection combining chi-square tests and mutual information with clinical relevance weighting; (5) final selection of 325 features per antibiotic model.

Critical SNP Exclusion:
A significant finding was the systematic exclusion of all single nucleotide polymorphisms during feature selection, resulting in final feature sets comprising exclusively antimicrobial resistance genes. This exclusion occurred due to the prioritization strategy favoring high clinical relevance and statistical significance, potentially limiting model access to important resistance-associated sequence variants.

6.3 Machine Learning Implementation

Tree-Based Model Configuration:
XGBoost and LightGBM models employed binary logistic objectives with extensive hyperparameter optimization: n_estimators (100, 200, 500), max_depth (3, 5, 7), learning_rate (0.01, 0.1, 0.3). Class imbalance was addressed through scale_pos_weight parameters calculated as the ratio of negative to positive training examples. Geographic-temporal cross-validation with 5 folds ensured that related strains from the same location-year combinations were not split between training and validation sets.

Deep Learning Architecture:
DNABERT-2 implementation used a custom DNA-optimized BERT architecture with 256 hidden units, 6 transformer layers, and 8 attention heads for computational efficiency. Input sequences were limited to 512 tokens with nucleotide-level tokenization (vocab_size=9: A,T,G,C,N plus special tokens). Training employed AdamW optimization with learning rate 2e-5, early stopping based on validation F1-scores, and weighted CrossEntropyLoss to address class imbalance.

Cross-Validation Strategy:
Geographic-temporal cross-validation prevented data leakage by ensuring that isolates from the same location-year combinations remained together during train-validation splits. This approach addresses the challenge of phylogenetic relatedness in bacterial genomic data while providing realistic performance estimates for temporal generalization assessment.

6.4 Statistical Analysis and Evaluation Framework

Performance Metrics:
Primary evaluation used F1-scores with ≥0.85 threshold representing clinical-grade performance. Secondary metrics included balanced accuracy, precision, recall, and area under the receiver operating characteristic curve. All metrics were calculated on the temporally independent test set to provide unbiased performance estimates.

Interpretability Analysis:
SHAP (SHapley Additive exPlanations) analysis provided mechanistic interpretation for tree-based models through:

- global feature importance ranking using mean absolute SHAP values
- individual prediction explanations for resistance mechanism identification
- biological validation through comparison with established resistance databases and literature.

6.5 Computational Infrastructure

Hardware and Software:
Model training utilized a MacBook Pro M3 Pro (18GB RAM) for local testing and a dedicated Hetzner server(32 cores CPU, 128GB RAM, 1TB SSD). Software implementation employed Python 3.9, Snakemake 7.32 for workflow orchestration, scikit-learn for machine learning algorithms, XGBoost and LightGBM libraries, and PyTorch for deep learning implementation.

Reproducibility Framework:
Complete computational reproducibility was ensured through:

- conda environment specifications for all software dependencies
- version-controlled Snakemake workflows with explicit software versions
- automated result validation and collection scripts
- comprehensive logging of all hyperparameters and model configurations.

7. References

1. Abdollahi-Arpanahi, R., Gianola, D. and Peñagaricano, F. (2020) "Deep learning versus parametric and ensemble methods for genomic prediction of complex phenotypes," *Genetics Selection Evolution*, 52(1), p. 12. Available at: https://doi.org/10.1186/s12711-020-00531-z.
2. Ali, T., Ahmed, S. and Aslam, M. (2023) "Artificial Intelligence for Antimicrobial Resistance Prediction: Challenges and Opportunities towards Practical Implementation," *Antibiotics*, 12(3), p. 523. Available at: https://doi.org/10.3390/antibiotics12030523.
3. "Antimicrobial Resistance, Hypervirulent Klebsiella pneumoniae - Global situation" (no date). Available at: https://www.who.int/emergencies/disease-outbreak-news/item/2024-DON527.
4. Argimón, S. *et al.* (2021) "Rapid Genomic Characterization and Global Surveillance of *Klebsiella* Using Pathogenwatch," *Clinical Infectious Diseases*, 73(Supplement_4), pp. S325–S335. Available at: https://doi.org/10.1093/cid/ciab784.
5. Aytan-Aktug, D. *et al.* (2020) "Prediction of Acquired Antimicrobial Resistance for Multiple Bacterial Species Using Neural Networks," *mSystems*. Edited by C.W. Marshall, 5(1), p. 10.1128/msystems.00774-19. Available at: https://doi.org/10.1128/msystems.00774-19.
6. Bakri, M.M. (2022) "Study of Plasmid-Mediated Quinolone Resistance in Klebsiella pneumoniae: Relation to Extended-Spectrum Beta-Lactamases," *Journal of Pure and Applied Microbiology*, 16(2), pp. 1103–1110. Available at: https://doi.org/10.22207/JPAM.16.2.36.

7. Bartoszewicz, J.M., Seidel, A. and Renard, B.Y. (2021) "Interpretable detection of novel human viruses from genome sequencing data," *NAR Genomics and Bioinformatics*, 3(1), p. lqab004. Available at: https://doi.org/10.1093/nargab/lqab004.
8. Batisti Biffignandi, G. *et al.* (2024) "Optimising machine learning prediction of minimum inhibitory concentrations in Klebsiella pneumoniae," *Microbial Genomics*, 10(3). Available at: https://doi.org/10.1099/mgen.0.001222.
9. Bowers, J.R. *et al.* (2015) "Genomic Analysis of the Emergence and Rapid Global Dissemination of the Clonal Group 258 Klebsiella pneumoniae Pandemic," *PLOS ONE*. Edited by J.A. Bengoechea, 10(7), p. e0133727. Available at: https://doi.org/10.1371/journal.pone.0133727.
10. Budia-Silva, M. *et al.* (2024) "International and regional spread of carbapenem-resistant Klebsiella pneumoniae in Europe," *Nature Communications*, 15(1), p. 5092. Available at: https://doi.org/10.1038/s41467-024-49349-z.
11. Condorelli, C. *et al.* (2024) "Prediction of antimicrobial resistance of Klebsiella pneumoniae from genomic data through machine learning," *PLOS ONE*. Edited by M.K. Aworh, 19(9), p. e0309333. Available at: https://doi.org/10.1371/journal.pone.0309333.
12. Dalla-Torre, H. *et al.* (2025) "Nucleotide Transformer: building and evaluating robust foundation models for human genomics," *Nature Methods*, 22(2), pp. 287–297. Available at: https://doi.org/10.1038/s41592-024-02523-z.
13. Dickinson, Q. and Meyer, J.G. (2022) "Positional SHAP (PoSHAP) for Interpretation of machine learning models trained from biological sequences," *PLOS Computational Biology*. Edited by I. Ioshikhes, 18(1), p. e1009736. Available at: https://doi.org/10.1371/journal.pcbi.1009736.
14. Eraslan, G. *et al.* (2019) "Deep learning: new computational modelling techniques for genomics," *Nature Reviews Genetics*, 20(7), pp. 389–403. Available at: https://doi.org/10.1038/s41576-019-0122-6.
15. "European Centre for Disease Prevention and Control, World Health Organization, (2023) Antimicrobial resistance surveillance in Europe 2023 : 2021 data." (no date). Available at: https://data.europa.eu/doi/10.2900/63495.
16. Feng, H. (no date) "Benchmarking DNA Foundation Models for Genomic and Genetic Tasks."
17. Feng, L., Zhang, M. and Fan, Z. (2023) "Population genomic analysis of clinical ST15 Klebsiella pneumoniae strains in China," *Frontiers in Microbiology*, 14, p. 1272173. Available at: https://doi.org/10.3389/fmicb.2023.1272173.
18. Florensa, A.F. *et al.* (2022) "ResFinder – an open online resource for identification of antimicrobial resistance genes in next-generation sequencing data and prediction of phenotypes from genotypes," *Microbial Genomics*, 8(1). Available at: https://doi.org/10.1099/mgen.0.000748.
19. Gao, Y. *et al.* (2024) "Machine learning and feature extraction for rapid antimicrobial resistance prediction of Acinetobacter baumannii from whole-genome sequencing data," *Frontiers in Microbiology*, 14, p. 1320312. Available at: https://doi.org/10.3389/fmicb.2023.1320312.
20. Gorrie, C.L. *et al.* (2022) "Genomic dissection of Klebsiella pneumoniae infections in hospital patients reveals insights into an opportunistic pathogen," *Nature Communications*, 13(1), p. 3017. Available at: https://doi.org/10.1038/s41467-022-30717-6.
21. Gunasekaran, H. *et al.* (2021) "Analysis of DNA Sequence Classification Using CNN and Hybrid Models," *Computational and Mathematical Methods in Medicine*. Edited by J. Hemanth, 2021, pp. 1–12. Available at: https://doi.org/10.1155/2021/1835056.
22. Guo, Y. *et al.* (2021) "Gene-Based Testing of Interactions Using XGBoost in Genome-Wide Association Studies," *Frontiers in Cell and Developmental Biology*, 9, p. 801113. Available at: https://doi.org/10.3389/fcell.2021.801113.
23. Hawkey, J. *et al.* (2022) "ESBL plasmids in Klebsiella pneumoniae: diversity, transmission and contribution to infection burden in the hospital setting," *Genome Medicine*, 14(1), p. 97. Available at: https://doi.org/10.1186/s13073-022-01103-0.
24. Her, H.-L. and Wu, Y.-W. (2018) "A pan-genome-based machine learning approach for predicting antimicrobial resistance activities of the *Escherichia coli* strains," *Bioinformatics*, 34(13), pp. i89–i95. Available at: https://doi.org/10.1093/bioinformatics/bty276.
25. Holt, K.E. *et al.* (2015) "Genomic analysis of diversity, population structure, virulence, and antimicrobial resistance in *Klebsiella pneumoniae* , an urgent threat to public health," *Proceedings of the National Academy of Sciences*, 112(27). Available at: https://doi.org/10.1073/pnas.1501049112.

26. Hu, K. *et al.* (2024) “Assessing computational predictions of antimicrobial resistance phenotypes from microbial genomes,” *Briefings in Bioinformatics*, 25(3), p. bbae206. Available at: https://doi.org/10.1093/bib/bbae206.
27. Ji, Y. *et al.* (2021) “DNABERT: pre-trained Bidirectional Encoder Representations from Transformers model for DNA-language in genome,” *Bioinformatics*. Edited by J. Kelso, 37(15), pp. 2112–2120. Available at: https://doi.org/10.1093/bioinformatics/btab083.
28. Joe, H. and Kim, H.-G. (2024) “Multi-label classification with XGBoost for metabolic pathway prediction,” *BMC Bioinformatics*, 25(1), p. 52. Available at: https://doi.org/10.1186/s12859-024-05666-0.
29. Kim, J.I. *et al.* (2022) “Machine Learning for Antimicrobial Resistance Prediction: Current Practice, Limitations, and Clinical Perspective,” *Clinical Microbiology Reviews*, 35(3), pp. e00179-21. Available at: https://doi.org/10.1128/cmr.00179-21.
30. Koo, P.K. *et al.* (2021) “Global importance analysis: An interpretability method to quantify importance of genomic features in deep neural networks,” *PLOS Computational Biology*. Edited by R.D. Kouyos, 17(5), p. e1008925. Available at: https://doi.org/10.1371/journal.pcbi.1008925.
31. Köster, J. and Rahmann, S. (2012) “Snakemake—a scalable bioinformatics workflow engine,” *Bioinformatics*, 28(19), pp. 2520–2522. Available at: https://doi.org/10.1093/bioinformatics/bts480.
32. Lam, M.M.C. *et al.* (2021) “A genomic surveillance framework and genotyping tool for Klebsiella pneumoniae and its related species complex,” *Nature Communications*, 12(1), p. 4188. Available at: https://doi.org/10.1038/s41467-021-24448-3.
33. Li, Y. *et al.* (2024) “Artificial intelligence in predicting pathogenic microorganisms’ antimicrobial resistance: challenges, progress, and prospects,” *Frontiers in Cellular and Infection Microbiology*, 14, p. 1482186. Available at: https://doi.org/10.3389/fcimb.2024.1482186.
34. Lin, B., Yan, S. and Zhen, B. (2025) “A machine learning method for predicting molecular antimicrobial activity,” *Scientific Reports*, 15(1), p. 6559. Available at: https://doi.org/10.1038/s41598-025-91190-x.
35. Lin, X. (no date) “The Global and Regional Prevalence of Hospital-Acquired Carbapenem-Resistant Klebsiella pneumoniae Infection: A Systematic Review and Meta-analysis.” Available at: https://doi.org/10.1093/ofid/ofad649.
36. Liu, G. *et al.* (2024) “VirusPredictor: XGBoost-based software to predict virus-related sequences in human data,” *Bioinformatics*. Edited by P.L. Martelli, 40(4), p. btae192. Available at: https://doi.org/10.1093/bioinformatics/btae192.
37. Lyu, J.-W. *et al.* (2023) “Rapid Prediction of Multidrug-Resistant Klebsiella pneumoniae through Deep Learning Analysis of SERS Spectra,” *Microbiology Spectrum*. Edited by J.L. Jacobs, 11(2), pp. e04126-22. Available at: https://doi.org/10.1128/spectrum.04126-22.
38. Macesic, N. *et al.* (2020) “Predicting Phenotypic Polymyxin Resistance in Klebsiella pneumoniae through Machine Learning Analysis of Genomic Data,” *mSystems*. Edited by Y.-L. Yang, 5(3), p. 10.1128/msystems.00656-19. Available at: https://doi.org/10.1128/msystems.00656-19.
39. Navon-Venezia, S., Kondratyeva, K. and Carattoli, A. (2017) “Klebsiella pneumoniae: a major worldwide source and shuttle for antibiotic resistance,” *FEMS Microbiology Reviews*, 41(3), pp. 252–275. Available at: https://doi.org/10.1093/femsre/fux013.
40. Nguembang Fadja, A. *et al.* (2021) “Identification of natural selection in genomic data with deep convolutional neural network,” *BioData Mining*, 14(1), p. 51. Available at: https://doi.org/10.1186/s13040-021-00280-9.
41. Nguyen, M. *et al.* (2019) “Using Machine Learning To Predict Antimicrobial MICs and Associated Genomic Features for Nontyphoidal *Salmonella*,” *Journal of Clinical Microbiology*. Edited by D.J. Diekema, 57(2), pp. e01260-18. Available at: https://doi.org/10.1128/JCM.01260-18.
42. Nsubuga, M. *et al.* (2024) “Generalizability of machine learning in predicting antimicrobial resistance in E. coli: a multi-country case study in Africa,” *BMC Genomics*, 25(1), p. 287. Available at: https://doi.org/10.1186/s12864-024-10214-4.
43. Pei, N. *et al.* (2022) “Large-Scale Genomic Epidemiology of Klebsiella pneumoniae Identified Clone Divergence with Hypervirulent Plus Antimicrobial-Resistant Characteristics Causing Within-Ward Strain Transmissions,” *Microbiology Spectrum*. Edited by H. Wang, 10(2), pp. e02698-21. Available at: https://doi.org/10.1128/spectrum.02698-21.

44. Raffelsberger, N. *et al.* (2023) “Community carriage of ESBL-producing *Escherichia coli* and *Klebsiella pneumoniae* : a cross-sectional study of risk factors and comparative genomics of carriage and clinical isolates,” *mSphere*. Edited by M. Castanheira, 8(4), pp. e00025-23. Available at: https://doi.org/10.1128/msphere.00025-23.
45. Ren, Y. *et al.* (2022) “Prediction of antimicrobial resistance based on whole-genome sequencing and machine learning,” *Bioinformatics*. Edited by I. Birol, 38(2), pp. 325–334. Available at: https://doi.org/10.1093/bioinformatics/btab681.
46. Roberts, D.R. *et al.* (2017) “Cross‐validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure,” *Ecography*, 40(8), pp. 913–929. Available at: https://doi.org/10.1111/ecog.02881.
47. Russo, T.A. and Marr, C.M. (2019) “Hypervirulent Klebsiella pneumoniae,” *Clinical Microbiology Reviews*, 32(3), pp. e00001-19. Available at: https://doi.org/10.1128/CMR.00001-19.
48. Sparta, B. *et al.* (2024) “Binomial models uncover biological variation during feature selection of droplet-based single-cell RNA sequencing,” *PLOS Computational Biology*. Edited by J. Fan, 20(9), p. e1012386. Available at: https://doi.org/10.1371/journal.pcbi.1012386.
49. Spooner, A. *et al.* (2025) “Benchmarking ensemble machine learning algorithms for multi-class, multi-omics data integration in clinical outcome prediction,” *Briefings in Bioinformatics*, 26(2), p. bbaf116. Available at: https://doi.org/10.1093/bib/bbaf116.
50. Su, M., Satola, S.W. and Read, T.D. (2019) “Genome-Based Prediction of Bacterial Antibiotic Resistance,” *Journal of Clinical Microbiology*. Edited by A.J. McAdam, 57(3), pp. e01405-18. Available at: https://doi.org/10.1128/JCM.01405-18.
51. Tang, R. *et al.* (2022) “Machine learning in predicting antimicrobial resistance: a systematic review and meta-analysis,” *International Journal of Antimicrobial Agents*, 60(5–6), p. 106684. Available at: https://doi.org/10.1016/j.ijantimicag.2022.106684.
52. Vihta, K.-D. *et al.* (2024) “Predicting future hospital antimicrobial resistance prevalence using machine learning,” *Communications Medicine*, 4(1), p. 197. Available at: https://doi.org/10.1038/s43856-024-00606-8.
53. Watson, D.S. (2022) “Interpretable machine learning for genomics,” *Human Genetics*, 141(9), pp. 1499–1513. Available at: https://doi.org/10.1007/s00439-021-02387-9.
54. Wong, J.L.C. *et al.* (2019) “OmpK36-mediated Carbapenem resistance attenuates ST258 Klebsiella pneumoniae in vivo,” *Nature Communications*, 10(1), p. 3957. Available at: https://doi.org/10.1038/s41467-019-11756-y.
55. Wyres, K.L., Lam, M.M.C. and Holt, K.E. (2020) “Population genomics of Klebsiella pneumoniae,” *Nature Reviews Microbiology*, 18(6), pp. 344–359. Available at: https://doi.org/10.1038/s41579-019-0315-1.
56. Xu, L., Sun, X. and Ma, X. (2017) “Systematic review and meta-analysis of mortality of patients infected with carbapenem-resistant Klebsiella pneumoniae,” *Annals of Clinical Microbiology and Antimicrobials*, 16(1), p. 18. Available at: https://doi.org/10.1186/s12941-017-0191-3.
57. Zhang, C., Lei, X. and Liu, L. (2021) “Predicting Metabolite–Disease Associations Based on LightGBM Model,” *Frontiers in Genetics*, 12, p. 660275. Available at: https://doi.org/10.3389/fgene.2021.660275.
58. Zhang, J. *et al.* (2019) “LightGBM: An Effective and Scalable Algorithm for Prediction of Chemical Toxicity–Application to the Tox21 and Mutagenicity Data Sets,” *Journal of Chemical Information and Modeling*, 59(10), pp. 4150–4158. Available at: https://doi.org/10.1021/acs.jcim.9b00633.
59. Zhang, Y. *et al.* (2021) “Robustifying genomic classifiers to batch effects via ensemble learning,” *Bioinformatics*. Edited by K. Janet, 37(11), pp. 1521–1527. Available at: https://doi.org/10.1093/bioinformatics/btaa986.
60. Zhou, Z. *et al.* (2024) “DNABERT-2: Efficient Foundation Model and Benchmark For Multi-Species Genome.” arXiv. Available at: https://doi.org/10.48550/arXiv.2306.15006.

8. Appendix

Code: https://github.com/NasirNesirli/kleb-amr-project
Data: https://www.ncbi.nlm.nih.gov/pathogens/isolates/#taxgroup_name:%22Klebsiella%20pneumoniae%22